

Express Riddler

29 April 2022

Riddle:

There are many fractions with a numerator of 1 whose decimal expansions don't go on to infinitely many decimal places. For example, $1/4$ is equivalent to the decimal 0.25, and $1/500$ is equivalent to 0.002. However, the decimal expansion of $1/3$ is $0.33333\dots$, a decimal that never terminates.

If you were to add up all these numbers—fractions with a numerator of 1 whose decimal expansions don't go on forever—what would be the sum? (Note: Before you ask, let's include the fraction $1/1$ in this group.)

Solution:

The series in question is the sum of the inverses of all numbers with only 2 or 5 as factors:

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{5} + \frac{1}{8} + \frac{1}{10} + \frac{1}{16} + \frac{1}{20} + \frac{1}{25} \dots$$

This can be rewritten as the products of two series, separated into the powers of 2 and 5:

$$\left(1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} \dots\right) \left(1 + \frac{1}{5} + \frac{1}{25} + \frac{1}{125} \dots\right)$$

Each of these two series is geometric and has a finite sum because of the result:

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

when $|r| < 1$. In this case,

$$\begin{aligned} \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n &= 2 \\ \sum_{n=0}^{\infty} \left(\frac{1}{5}\right)^n &= \frac{5}{4} \end{aligned}$$

So the series simplifies to $2(5/4)$, and the solution is $\boxed{5/2}$.